AP ICET 2014

Analytical Ability

Instructions [1 - 20]

In questions a question is followed by data in the form of two statements labelled as I and II. You must decide whether the data given in the statements are sufficient to answer the questions. Using the data make an appropriate choice from (a) to (d) as per the following guidelines :

1. **What is the positive integer n not exceeding 180?**
   I. n is divisible by 7.
   II. n is divisible by 13.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

2. **If ABCD is a square and is a point on BC, then whatis the area (in square units) of AECD?**
   I. BE = 6
   II. BE : EC = 1 : 2

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

3. **What is the shape of the play ground ?**
   I. The perimeter of the play ground is 440 m
   II. The area of the ground is 15400 sq. m.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

4. **What is the remainder when n is divided by 8 ?**
I. The digit in units place of n is 8.
II. n is the product of eight consecutive positive integers.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

5. **What is the greatest common divisor of numbers a and b ?**
I. The least common multiple of a and b is ab.
II. a + b = 15.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

6. **What is the average of a, b, c and 5?**
I. 5(a + b + c) + 4 = 45.
II. a + b = c + d

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

7. **What is the value of** $\frac{x^2}{y^2} + \frac{y^2}{z^2}$**?**

   I. $\left(\frac{x}{y} + \frac{y}{z}\right)^2 = 100$

   II. x = 2z

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

8. **Is** $xy < 0$?
   I. $5 \mid x \mid +3 \mid y \mid= 0$
   II. $5 \mid x \mid= 3 \mid y \mid$

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

9. **How much time did A take to reach the destination ?**
   I. The ratio between the speeds of A and B is 3 : 4.
   II. B takes 36 minutes to reach the same destination.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

10. **What is the slope of straight line ?**
I. The straight line passes through the origin.
II. The straight line makes an angle 30° with the positive direction of the X-axis.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

11. **In the matrix** $A = \begin{bmatrix} -5 & 20 \\ 2 & x \end{bmatrix}$**, what is the value of the x?**

I. A is singular.

II. A is symmetric.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

12. **What is the value of a + b?**
I. a ≠ b
II. $a^2 - b^2 = a - b$

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

13. **Is the quadniateral a square ?**
    I. All the sides of the quadrilateral are of equal length.
    II. The diagonals of the quadrilateral are of equal length.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

14. **For positve integers x, y and z, is the product xyz even ?**
    I. x + y is odd.
    II. x + y + z is divisible by 7.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

15. **What is the monthly salary of A ?**
    I. A gets 15% more than B and B gets 10% less than C.
    II. C's monthly salary is ₹ 2,500,

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

16. **Among the real numbers a and b, is b a rational number ?**
I. a + b is a rational number.
II. a - b is a rational number.

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

17. **How many persons are there in the library ?**
I. If 3 persons leave the library, then the library has less than 8 persons.
II. If 3 persons enter the library, then it has more than 12 persons.

A if the statement I alone is sufficient to answer the question.

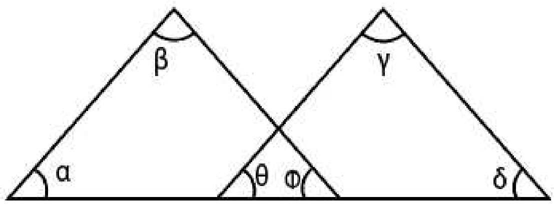
B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

18. **In the figure given below, what is the value $\alpha+\beta+\gamma+\delta$?**



I. $\alpha+\beta=\gamma+\delta$

II. $\theta+\phi=90°$

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

19. **How much is (x + y) : (x - y)?**
I. x : y = 3 : 2
II. x > 0, y > 0

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

20. **If p(x) is a polynomial, is (x - 2) a factor of $p(2x^2-1)$?**
I. x = 1 is a factor of p(x).
II. x - 7 is factor of p(x).

A if the statement I alone is sufficient to answer the question.

B if the statement II alone is sufficient to answer the question.

C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

Instructions [21 - 30]

In each of the questions numbered a sequence of numbers and letters that follow a definite pattern is given. Fach question has a blank space. This blank space 1 to be filled by the correct answer from one of the four given options to complete the sequence without breaking the pattern.

21. **7 : 49 :: .... : 63**

A 5

B 6

C 9

D 11

Answer: C

22. **81 : 64 :: .... : 9**

A 16

B 18

C 24

D 34

Answer: A

23. **AEF : BIJ :: .... : OUV**

A NOP

B MPQ

C NOQ

D NQR

Answer: D

24. **DRIVE : EIDRV :: BEGUM : ....**

A BGMEU

B MGBEU

C UEBGM

D BGMUE

Answer: B

25. **E x I : 5 x 9 :: ..... : 15 x 21**

A L x K

B K x L

C O x U

D U x O

Answer: C

26. **ANT : CPV :: ..... : DQZ**

A BOX

B BRB

C FSB

D FTB

Answer: A

27. **BCEH, ..., DGKP, EINT**

A CDJG

B CEJK

C CFIM

D CEHL

Answer: D

28. **K 11 M, ..., G 15 I, E 17 G**

A I 13 K

B I 14 J

C I 12 J

D I 13 M

Answer: A

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29. **HOSPITAL : PATIENTS :: SCHOOL : ....**

A TEACHERS

B CLASS ROOMS

C STUDENTS

D BLACK BOARDS

Answer: C

30. **If the letters D and E are removed from the English alphabet, then ihie fourth letter is**

A F

B C

C G

D H

Answer: A

Instructions [31 - 35]

pick the odd thing out :

31.

A 147

B 125

C 103

D 84

Answer: B

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32.

A (3, 4, 5)

B (5, 12, 13)

C (6, 8, 10)

D (10, 12, 15)

Answer: D

33.

A April

B May

C November

D September

Answer: B

A $\frac{19}{15}$

B $\frac{13}{11}$

C $\frac{7}{5}$

D $\frac{3}{2}$

Answer: A

34.

A C 4 E

B G 8 I

C L 15 N

D T 21 V

Answer: C

Instructions [35 - 44]

Each of the questions follow a definite pattern. Observe the same and fill in the blanks with suitable answers.

35. $111\frac{1}{9}, 125, 142\frac{6}{7}, ..., 200, 250$

A $166\frac{2}{3}$

B $178\frac{4}{7}$

C $181\frac{2}{5}$

D $192\frac{3}{7}$

Answer: A

36. **0, 2, 3, 5, 8, 10, 15, ..., 24, 26, 35**

A 19

B 18

C 17

D 16

Answer: C

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37. $\left\{\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}\right\}, \left\{\frac{1}{12}, \frac{1}{15}, \frac{1}{18}, \frac{1}{21}\right\}, \left\{\frac{1}{25}, \frac{1}{29}, \frac{1}{33}, \frac{1}{37}\right\}, \left\{\frac{1}{42}, \frac{1}{47}, ..., \frac{1}{57}\right\}$

A $\frac{1}{50}$

B $\frac{1}{51}$

C $\frac{1}{52}$

D $\frac{1}{53}$

Answer: C

38. **5, 11, 21, 43, 85, ...**

A 181

B 180

C 171

D 170

Answer: C

39. **75, 105, 165, 195, ..., 285**

A 255

B 235

C 225

D 215

Answer: A

40. $\frac{3}{5}, \frac{5}{9}, \frac{9}{13}, \frac{13}{17}, ..., \frac{27}{33}$

A $\frac{11}{13}$

B $\frac{17}{27}$

C $\frac{15}{19}$

D $\frac{21}{32}$

Answer: B

41. **(1, Z), (8, Y), (27, X), (125, W), ...**

A (243, U)

B (243, V)

C (343, V)

D (343, U)

Answer: C

42. **AEI, CGK, ..., GKO, IMQ**

A EJN

B ENJ

C EIM

D EMI

Answer: C

43. If $\{a_n\}_{n-1}^{\infty}$ is such that $a_1 = a_2 = 1$ and $a_k = a_1 + a_2 + ... + a_{k-1}$ for $k \geq 3$, then $a_7 =$

A 16

B 32

C 64

D 128

Answer: B

44. Then n^{th} term in the sequence
1, -2, 3, -4, 5, -6, 7, -8, ... is

A $(-1)^n n$

B -n

C n

D $(-1)^{n-1}.n$

Answer: D

Instructions [45 - 47]

An automobile company produces four types of vehicles (Cars, Motor bikes, Scooters and Mopeds) at different branches in the country. The production at these units from 2007 to 2012 are given in the table below. Answer the questions using the table.

	2007	2008	2009	2010	2011	2012
Cars	3600	6300	8100	10800	16200	19800
Motor bikes	7000	12250	15750	21000	31500	38500
Scooters	8000	14000	18000	24000	36000	44000
Mopeds	9000	15750	20250	27000	40500	49500

45. The ratio of the number of Cars produced in 2008 to the number of Scooters produced in 2011 is

A 37 : 40

B 27 : 40

C 17 : 40

D 7 : 40

Answer: D

46. **In which year the total number of the four types of vehicles produced was 62100 ?**

A 2007

B 2008

C 2009

D 2010

Answer: C

47. **If k : 1 is the ratio of the number of Scooters produced in the year 2011 to the number of scooters produced in 2007, then k =**

A 3

B 4

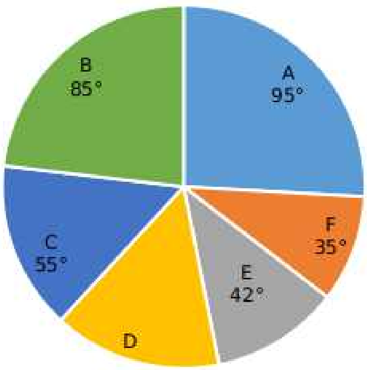
C $\frac{9}{2}$

D $\frac{2}{9}$

Answer: C

Instructions [48 - 52]

The expenditure under six heads A, B, C, D, E and F in an year are as given in the following Pie diagram. Answer the questions using the diagram.



48. **If the total expenditure in an year is ₹ 54,00,000, then the expenditure (in rupees) under the head E in that year is**

A 3,60,000

B 4,20,000

C 6,30,000

D 7,20,000

Answer: C

49. **If the expenditure under the heads A and B together 1s ₹ 18,00,000 in an year, then the expenditure under the head D in that year is**

A ₹4,20,000

B ₹4,80,000

C ₹5,50,000

D ₹8,50,000

Answer: B

50. **If the difference in the expenditure under the heads A and B in an year is ₹ 2.5 lakhs, then the total expenditure (in lakhs of rupees) in that year is**

A 90

B 81

C 72

D 63

Answer: A

51. **If the expenditure under the head E in an year is ₹ 3.5 lakhs, then the expenditure (in lakhs of rupees) under the head D in that year is**

A 3

B 3.5

C 4

D 4.5

Answer: C

52. **In any year the expenditure under the heads C and F together is equal to**

A half of the expenditure on B and together.

B double the expenditure on C.

C the expenditure on D and E.

D the expenditure on A.

Answer: C

Instructions [53 - 54]

Given that A = {n : n prime, $1 \leq n \leq 20$}
B = {n : n odd, $1 \leq n \leq 20$}
and C = {n : n square, $1 \leq n \leq 20$}.
Using this answer the questions

53. $B \cap C =$

A {1, 4, 9, 16}

B (1, 3, 5, 7, 9, 11, 13, 15, 17, 19}

C ϕ

D {1, 9}

Answer: D

54. **The number of integers between 1 and 20 which do not lie in $A \cup B \cup C$ is**

A 8

B 7

C 6

D 5

Answer: B

Instructions [55 - 59]

The letters A, B, C ,....... Z of English alphabet are numbered 1, 2, 3, ..., 26 respectively. A code is designed by shifting r^{th} letter to $(14 - r)^{th}$ letter if $1 \leq r \leq 13$ and s^{th} letter to $(40 - s)^{th}$ letter if $14 \leq s \leq 26$. The reverse process is used for decoding. Using this answer questions.

55. **The code word for HYDERABAD is**

A FOJIVMLMJ

B FJOIVMJML

C FIJOVMLMJ

D FIOJMVLMJ

Answer: A

56. **The code word for WARANGAL is**

A QMUMZBGM

B QMVMZGMB

C QVMZMBGM

D QVMZMGBM

Answer: B

57. **Which word is coded as IPKFMZGI ?**

A ENTRANCE

B ELEGANCE

C EXCHANGE

D EMPATHIE

Answer: C

58. **Which word is coded as UIBIKTEYZ ?**

A SEDUCTION

B SELECTION

C SUGGESTED

D SHOCKINGS

Answer: B

59. **The code word for TIRUPATI is**

A TEMXVSTE

B TESVXMTE

C TESVMXTE

D TEVSXMTE

Answer: D

60. **If TEACHER is coded as UFBDIFS, then the code word for PARENT is**

A QSBOFU

B QBSFOU

C RBSENU

D QRAESU

Answer: B

61. **If BOMBAY is coded as OBZONL, then the code for DELHI is**

A QRYVU

B QYRUV

C QRYUV

D RQYUV

Answer: C

62. **If COMMERCE is coded as DONNESDE, then the code for BIOLOGY is**

A CJPMPHZ

B CIOMOHZ

C CIOOMHZ

D CIOMOZH

Answer: B

63. **If FAILURE is coded as EZHKTQD, then the code for SUCCESS is**

A RSDDFRR

B RTDDFRR

C RTBBDRR

D RTCBRDR

Answer: C

64. **If TRLANGLE is coded as USJBOHMF, then the code word for SQUARE is**

A TRUBSF

B TRVSBF

C TRUSBF

D TRVBSF

Answer: D

65. **If 26^{th} January of a non-leap year falls on a Sunday, the day on which 15^{th} August of that year falls is**

A Sunday

B Wednesday

C Friday

D Saturday

Answer: C

66. **What is the angle between the two hands of a clock when the time is 5.15 a.m. ?**

A $72\frac{1}{2}^{\circ}$

B $67\frac{1}{2}^{\circ}$

C 64°

D $58\frac{1}{2}^{\circ}$

Answer: B

67. **A clock strikes once at 1 O'clock, twice at 2 O'clock and so on. The total number of strikes it makes in a day is**

A 78

B 112

C 132

D 156

Answer: D

68. **D is the only son of his father C. A is B's brother and B is C’s sister. How is D related to A?**

A Sister

B Brother

C Niece

D Brother's son

Answer: D

69. **A meeting 18 scheduled at 11.00 am for which a person P who is away at 100 kms from the venue has to attend. If P starts at 9.45 a.m. in a car which moves with a speed of 60 kmph, then the P is late to the meeting by how many minutes ?**

A 5

B 15

C 25

D 35

Answer: C

70. **If t_1 is the time elapsed between 11.10 am to 3.50 pm: and if t_2 is the time elapsed between 10.15 am to 4.05 pm, then $t_1 : t_2 =$**

A 5 : 4

B 2 : 3

C 4 : 5

D 3 : 2

Answer: C

71. **A, B, C, D and E sit around a table such that A is between B and C and is left to B : D is to the right of B; and E is between C and D. Then the person to the immediate left of C is**

A D

B B

C A

D E

Answer: D

72. Given that $a * b = \frac{a^2+b^2}{ab}$ and $a \triangle b = \frac{a^2}{b}$ for any real numbers a and b. If $x * y = 2 \triangle 2$, then x =

A y

B $\frac{y}{2}$

C 2y

D $3^{\frac{y}{2}}$

Answer: A

73. For real numbers a and b, if $a \circ b = (ab)^{\frac{1}{5}}$, then $(243) \circ (16807) =$

A 31

B 29

C 22

D 21

Answer: D

74. If a □ b = $(a + b - 1)^2 - 1$, then (1□2)□(3□ 3) =

A 576

B 625

C 675

D 676

Answer: C

Mathematical Ability

75. $\left(a^{\frac{1}{z-x}}\right)^{\frac{1}{z-y}} . \left(a^{\frac{1}{x-y}}\right)^{\frac{1}{x-z}} . \left(a^{\frac{1}{y-z}}\right)^{\frac{1}{y-x}} =$

A a

B 0

C xyz

D 1

Answer: D

Explanation:

$$\left(a^{\frac{1}{z-x}}\right)^{\frac{1}{z-y}} \cdot \left(a^{\frac{1}{x-y}}\right)^{\frac{1}{x-z}} \cdot \left(a^{\frac{1}{y-z}}\right)^{\frac{1}{y-x}} =$$

$$= a^{\left(\frac{1}{(x-z)(y-z)}\right)} \cdot a^{\left(\frac{1}{(x-y)(x-z)}\right)} \cdot a^{\left(-\frac{1}{(y-z)(x-y)}\right)}$$

$$= a^{\left(\frac{1}{(x-z)(y-z)} + \frac{1}{(x-y)(x-z)} - \frac{1}{(y-z)(x-y)}\right)}$$

Adding up all the terms in the powers yields 0 as the answer.
Hence, a raised to 0 = 1

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76. **If** $\left(\sqrt{\frac{3}{5}}\right)^{a} = \left(\sqrt{\frac{625}{81}}\right)^{\frac{a+3}{2}}$ **, then a =**

A 2

B 1

C -1

D -2

Answer: D

Explanation:

The RHS of the equation can be modified as $\left(\sqrt{\frac{81}{625}}\right)^{-\frac{(a+3)}{2}} = \left(\sqrt{\frac{3}{5}}\right)^{4\cdot\left(-\frac{(a+3)}{2}\right)}$

As the bases are equal on both sides, respective powers can be equated:
a = -2*(a+3)
a = -2

77. **In a mixture of 35 litres the ratio of milk and water is 4 : 1. If one litre of water is added to the mixture the ratio ofmilk and water in the new mixture is**

A 2 : 7

B 7 : 2

C 4 : 3

D 2 : 1

Answer: B

Explanation:
The amount of milk present in the solution at start = (4/5)*35 = 28 liters

Amount of water present at the start = 35 - 28 = 7 liters
Now, one liter water is added and hence, total water present in solution = 8 liters.
Hence, ratio of milk and water in the mixture now = 28 : 8 = 7 : 2

78. **The salaries of two persons are in the ratio 4: 7. Both spend 80% of their salaries and save the rest. The ratio of their savings is**

A 8 : 2

B 7 : 5

C 5 : 3

D 4 : 7

Answer: D

Explanation:
Let the salaries be 4x and 7x.
As both spend 80% of their salaries and save 20%, the amounts saved are 0.2*4x = 0.8x and 0.2*7x = 1.4x
The ratios of savings = 0.8:1.4 = 4:7

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79. **If** $(\sqrt{2})^{x+5} = (\sqrt[4]{2})^{2x^2-2}$**, then a value of** $(x^2 - 1)$ **is**

A 2

B 4

C 6

D 8

Answer: D

Explanation:
The RHS of the equation can be re written as $(\sqrt{2})^{\frac{2x^2-2}{2}} = (\sqrt{2})^{x^2-1}$
As the bases in both LHS and RHS are now same, the powers can be equated.

$x + 5 = x^2 - 1$
solving this equation gives us x = 3 or x = -2
For the desired value, we get either 8 or 3. Option having 8 is present, and hence, the answer.

80. $|\sqrt{10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}}| + |\sqrt{10 - 2\sqrt{6} - 2\sqrt{10} + 2\sqrt{15}}| =$

A $2(\sqrt{3} + \sqrt{5})$

B $2\sqrt{3}$

C $2\sqrt{5}$

D $2\sqrt{10}$

Answer: A

Explanation:
Firstly, we should know the identities:
$(a+b+c)^2 = a^2+b^2+c^2+2ab+2ac+2bc$
$(a-b-c)^2 = a^2+b^2+c^2-2ab-2ac+2bc$

In the question, the first expression inside the square root can be rewritten as:
$10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15} = 2+3+5+2\sqrt{2}\sqrt{3}+2\sqrt{2}\sqrt{5}+2\sqrt{3}\sqrt{5}$=
$(\sqrt{2}+\sqrt{3}+\sqrt{5})^2$
Hence, the first modulus will return the net value $=\sqrt{2}+\sqrt{3}+\sqrt{5}$

Similarly, the expression in the second square root can be simplified as $=(\sqrt{2}-\sqrt{3}-\sqrt{5})^2$
As sqrt(3) and sqrt(5) are greater than sqrt(2), the modulus will change the sign of the expression finally and it will becomes $=-\sqrt{2}+\sqrt{3}+\sqrt{5}$

Hence, the resultant value will be option A

81. **The least value of k such that $315 \times k$ is a perfect square is**

A 35

B 31

C 21

D 15

Answer: A

Explanation:
315 x k = 3 x 3 x 5 x 7 x k
To make the LHS a perfect square, k should definitely have at least one power of 5 and 7.
For the least value of k, we should have only the minimum requirements i.e. k = 5 x 7
hence, value of k must be 35.

82. **Which among the following numbers leaves remainders 1, 2 and 2 respectively when divided by 2, 3 and 7?**

A 130

B 68

C 65

D 57

Answer: C

Explanation:
The best approach in these questions is to evaluate the options and arrive at the required answer by some trial and error.
As we can see, the desired number has to be an odd number because 2 cannot divide it completely. Hence, options A and B get eliminated quickly.
The required conditions get satisfied with option C.

The theoretical approach for such questions is long and not advisable when options are present.

83. **The L.C.M. of two integers is 144 and their G.C.D is 12. If one of the integers is 36, then the other integer is**

A 18

B 24

C 48

D 432

Answer: C

Explanation:
Given, LCM = 144

GCD = HCF = 12

One of the integer = 36

Let the other integer = a

We know that,

Product of 2 integers = HCF x LCM

$\Rightarrow$ 36 x a = 144 x 12

$\Rightarrow$ a = 48

$\therefore$ The other integer = 48

Hence, the correct answer is Option C

84. **The least number that is to be subtracted from 2580 so that it leaves a remainder 4 when divided by 9, 11 and 13 is**

A 1

B 2

C 3

D 4

Answer: B

Explanation:
The question asks for a number which will leave remainder 4 whether it is divided by 9 or 11 or 13. The number has to be less than 2580 and largest possible in this range.

This means that the desired number will be of the form = k*(LCM of 9,11,13) + 4
where K means any multiple of the LCM of 9,11,13 and should result in a number less than 2580
LCM if 9, 11, 13 is 1287. For k = 2, we get the desired number as = 2574 + 4 = 2578
Hence, 2 must be subtracted from 2580 to get desired number.

85. **Three numbers are in the ratio 1 : 2 : 3 and the sum of their squares is 504. The largest of the numbers is**

A 6

B 12

C 18

D 24

Answer: C

Explanation:
Let the 3 numbers be a, 2a, 3a respectively.

According to the problem,

$a^2 + 4a^2 + 9a^2 = 504$

$\Rightarrow\ 14a^2 = 504$

$\Rightarrow\ a^2 = 36$

$\Rightarrow\ a = 6$

$\therefore$ The largest of the numbers = 3a = 18

Hence, the correct answer is Option C

86. **The ascending order of the fractions :** $\frac{5}{7}, \frac{6}{8}, \frac{9}{11}, \frac{11}{14}$ **is**

A $\frac{5}{7}, \frac{6}{8}, \frac{9}{11}, \frac{11}{14}$

B $\frac{5}{7}, \frac{9}{11}, \frac{6}{8}, \frac{11}{14}$

C $\frac{5}{7}, \frac{11}{14}, \frac{9}{11}, \frac{6}{8}$

D $\frac{5}{7}, \frac{6}{8}, \frac{11}{14}, \frac{9}{11}$

Answer: D

Explanation:
By observation, we can see that (5/7) is less than (11/14) as 5/7 is equal to (10/14)
Comparing 5/7 to 6/8 we can see that (5/7) is less than (6/8).

Lastly, we can compare (9/11) with (11/14). The approx value of the first is 0.81 while the value of the second one is approx. <0.8
Hence, the correct ascending order is the one given in the answer option.

87. **The persons A, B, C share a property in such a way that A and B get** $\frac{3}{7}th$ **and** $\frac{5}{14}th$ **and C getting the rest. The person or persons who get the least property.**

A C

B B

C A and B

D A and C

Answer: A

Explanation:
A gets (3/7)th of the property means that he gets (6/14)th of the property.
B gets (5/14)th of the property. This leaves (3/14)th of the property of C and hence, least amount is received by C.

88. **The descending order of** $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$ **is**

A $\sqrt[4]{10}, \sqrt{3}, \sqrt[3]{6}$

B $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$

C $\sqrt{3},\ \sqrt[3]{6},\ \sqrt[4]{10}$

D $\sqrt[3]{6},\ \sqrt[4]{10}, \sqrt{3}$

Answer: D

Explanation:
We raise all the terms to an equal power. Let's take the 12th power of each term.
We get $6^4,\ 10^3\ and\ 3^6$

The value of the 4th power of 6 is easily over 1000, which is nothing but cube of 10. And lastly 6th power of 3 is 729.
Hence, the descending order of the terms would be the answer option.

89. **In a face to face election the winner got 65% of votes and won by a margin of 12000 votes. The total votes polled (in lakhs) is**

A 4

B 0.4

C 0.04

D 0.004

Answer: B

Explanation:
Let total votes polled be V.
The winner got 65% votes and won by margin of 12000 votes. Means the rival got 35% votes.
hence, 0.65V - 0.35V = 12000 which gives us V = 12000/0.3 = 40,000 i.e. 0.4 lakh

90. **In a library 23% of the books are in Arts. 30% in Commerce, 35% in Science and the rest are in Telugu language. If there are 1440 books in Telugu language, the number of books in Arts is**

A 2760

B 3000

C 3600

D 4200

Answer: A

Explanation:
Let the total number of books be T.
we can see that 12% of the books are in Telegu language.
Hence, 0.12T = 1440 which gives us T = 144000/12 = 12000
Now, number of arts books = 0.23T = 120*23 = 2760

91. **A person bought a pen and sold it for a loss of 10%. If he had bought it for 20% less and sold it for ₹ 44 more than earlier sale price he would have made a profit of 40%. The cost price of the pen is (in ₹)**

A 200

B 225

C 250

D 280

Answer: A

Explanation:
Let the original cost price be CP

When the pen is originally sold for a loss of 10%, then original SP = 0.9CP

Now, it is said that if the pen was purchased for 20% less i.e. if cost price had been 0.8CP and the pen was sold for Rs. 44 more than original selling price i.e. if the pen had been sold for (0.9CP + 44) rupees, the profit % would have been 40% i.e.

(0.9CP + 44) = 1.4*(0.8CP) which gives us 0.22CP = 44
Hence, CP = 200.

92. **If an article is sold at a profit of 15% instead of a profit of 9% the person gets ₹ 60 more. The cost price ofthe article (in rupees) is**

A 1200

B 1050

C 1000

D 800

Answer: C

Explanation:
Selling Price when 15% profit is to be made on cost price = SP1 = 1.15CP
Selling Price when 9% profit is to be made on cost price = SP2 = 1.09CP

It is given that SP1 - SP2 = 60 i.e. 1.15CP - 1.09CP = 60
Hence, 0.06CP = 60 which makes CP = 1000

93. **A and B started a business investing ₹ 10 lakhs and ₹ 15 lakhs respectively. After 6 months C joined them by investing ₹ 20 lakhs. If the profit at the end of the year is ₹ 5.6 lakhs, then the share of A in the profit (in lakhs of rupees) is**

A 1.6

B 2.4

C 3.2

D 4.8

Answer: A

Explanation:
Profits are always shared in the ratio of (capital*time) ratio.
Here, A makes an investment of Rs. 120 lakh-months, B makes an investment of 180 lakh-months and C makes an investment of 120 lakh-months.
The ratio of investments of A, B, C thus are: 120:180:120 = 2:3:2

Hence, the ratio of A in the year end profits would be (2/7)*(5.6 lakhs) = 1.6 lakhs

94. **In a joint business A, B and C invested capital in the ratio 5 : 6 : 8. At the end of the business they shared profits in the ratio 4 : 3 : 12.'The ratio of the number of months in which A, B and C kept, their capital is**

A 2 : 1 : 3

B 5 : 3 : 12

C 8 : 85 : 15

D 25 : 18 : 16

Answer: C

95. **Pipe A fills a tank in 8 hours while pipe B empties the full tank in 10 hours. If both the pipes A and B are opened simultaneously the time taken (in hours) to fill the tank is**

A $33\frac{1}{2}$

B $36\frac{1}{2}$

C 40

D 42

Answer: C

Explanation:
Let the tank capacity be 80 liters.
Hence, the first pipe can fill the tank at 10 liters per hour and the draining pipe drains the tank at 8 liters per hour.

Both are opened together, so the net effect would be 10-8 = 2 liters of volume of liquid added per hour.
Hence, amount of time required to fill tnk completely = 80/2 = 40 hours

96. **Two pipes A and B can fill a tank in 10 hours and 15 hours respectively. If they are opened altemately for one hour each and if A is opened first, the time (in hours) required to fill the tank is**

A 10

B 11

C 12

D 13

Answer: C

Explanation:

Let the total capacity of the tank be 30 liters

Hence, the capacity of the first pipe is 3 liters/hour and the capacity of the second pipe is 2 liters per hour. Now, both the pipes are opened alternately for an hour each. Thus, in one round (or cycle) of 2 hours, net volume of tank filled = 3+2 = 5 liters
Hence, the tank will be completely filled in 30/5 = 6 rounds i.e. 12 hours

97. **If a man starts at A and walks at 5 kmph he will reach B late by 7 minutes. But if walks at 6 kmph he will reach B early by 5 minutes. The distance between A and B (in km) is**

A 4

B 5

C 6

D 7

Answer: C

Explanation:
Let the distance between A and B be d km and the ideal time needed to reach the destination be t hours

Now, in the first case, we can say that $\frac{d}{5} = t + \frac{7}{60}$
That is $t = \frac{d}{5} - \frac{7}{60}$

Now, in second case, we can say that $\frac{d}{6} = t - \frac{5}{60}$
That is $t = \frac{d}{6} + \frac{5}{60}$

Equating both the sides for t, we get d = 6 km

98. **A train of 270 metres long crosses a platform of 390 metres length in 33 seconds. The speed of the train (in kmph) is**

A 66

B 68

C 72

D 75

Answer: C

Explanation:
The total distance travelled by the train in 33 seconds = 270 + 390 meters = 660 meters

Speed of train (in m/s) = 660/33 = 20 m/s
Hence, speed in kmph = 20*(18/5) = 72 kmph

99. **Three persons A, B, C together can complete a work in 8 days where as A alone requires 24 days to complete the same work. The number of days-required for B and C together to complete the same work is**

A 18

B 16

C 12

D 10

Answer: C

Explanation:
Let the total amount of work be 24 units.
Hence, by given data, we can say that A alone completes 1 unit per day and A,B,C all together do 3 units of work per day.
Hence, we can say that B and C together do 2 units of work per day.

Thus, the amount of time required by only B and C together to do entire work = 24/2 = 12 days

00. **A man completes $\frac{4}{5}th$ of the work in $1\frac{1}{2}$ days. The number of hours required to complete the remaining work by him is**

A 6

B 9

C 7

D 8

Answer: B

Explanation:
The speed of doing work is assumed to be constant. Hence, direct variation concept is applicable here.

$$\frac{\left(Time\ to\ complete\ \left(\frac{4}{5}\right)th\ work\right)}{Time\ to\ complete\ \left(\frac{1}{5}\right)th\ work} = \frac{(1.5\ days)}{x}$$

Hence, x = (3/8)th day = 3*24/8 hours = 9 hours

01. **A circle is inscribed in an equilateral triangle. If the area of the circle is 462 cm ^{2}, then the perimeter (in cm) of the triangle is**

A 72

B 84

C 96

D 126

Answer: D

Explanation:
In an equilateral triangle, the incenter, orthocenter, median are all coincident.
Thus, the center of the inscribed circle is also the centroid of the triangle and the radius of the circle is the smaller division of the median which gets cut by the centroid. The centroid divides the median in the ratio of 2:1. Thus, we can say that the height of the triangle will be 3 times the radius of the inscribed circle.

Radius of the circle = sqrt (462*7/22) = $7\sqrt{3}$
Height of the triangle = $21\sqrt{3}$

Now, Height of triangle = $\frac{\sqrt{3}}{2} \cdot Side\ of\ triangle$
Hence, Side of triangle = 42 cm and thus, perimeter = 126 cm

02. **The area of a rectangular metal sheet is 60 sq.m. The sum of its length and diagonal is equal to 5 times its breadth, Then the difference (in metres) between length and breadth is**

A 4

B 5

C 6

D 7

Answer: D

Explanation:
Let the length , breadth, diagonal length be l, b and d respectively.
We know that d = $\sqrt{l^2 + b^2}$

We are given that :
$l + \sqrt{l^2 + b^2} = 5 \cdot b$
$\sqrt{l^2 + b^2} = 5 \cdot b - l$
Squaring both sides and then putting l*b = 60 (given in question), we get:
$b^2 = 25b^2 - 600$
$b = 5$
Hence, l = 12 and difference in length and breadth = 7

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03. **A cone of height 24 cm and radius of its base 6 cm is made up of clay. If that clay is reshaped in the form of a sphere, then the diameter of that sphere (in cms) is**

A 6

B 8

C 12

D 14

Answer: C

Explanation:
The volume of the clay available remains constant.
Hence, equating the volumes of the cone and the sphere:
$\frac{1}{3}\pi \cdot 6^2 \cdot 24 = \frac{4}{3}\pi \cdot R^3$

Solving this, we get R = 6, and hence, diameter = 12.

04. **The surface area of a sphere is same as the curved surface area of a right circular cylinder whose height and diameter are 12 cm each. Then the radius of the sphere (in cm) is**

A 3

B 4

C 5

D 6

Answer: D

Explanation:
Let radius of the sphere be R.

Hence, according to given data, we get
$4 \cdot \pi \cdot R^2 = 2 \cdot \pi \cdot 6 \cdot 12$
where, the 6 in RHS is the radius of the cylinder and 12 is the height

Solving the above equation, we get R = 6

05. **Let "s" be the surface area of a cube of edge 9 em. This cube is cut into smaller cubes of edge 3 cm each, If "S" is the sum of the surface areas of all the smaller cubes, then s : S =**

A 3 : 1

B 1 : 3

C 3 : 2

D 2 : 3

Answer: B

Explanation:
Number of smaller cubes obtained = (Volume of original cube)/(Volume of each of the smaller cube)
Hence, Number of smaller cubes obtained = (9*9*9)/(3*3*3) = 27

Now, surface area "s" of the original cube = 6*(9*9) sq. cm
Sum of Surface areas of all the smaller cubes, S = 27 * [6*(3*3)]

Hence, s:S = 1:3

06. **The number of revolutions made by a wheel of 4? cm diameter in travelling a distance of 1320 metres is**

A 300

B 400

C 500

D 1000

Answer: D

07. **The radius r of a right circular cylinder is the same as that of a sphere. If the volume of the sphere is twice that of the cylinder, then the height of the eylinder is**

A $\frac{r}{3}$

B $\frac{2r}{3}$

C $\frac{4r}{3}$

D $2r$

Answer: B

08. **The digit in the units place of the number 13^{400} is**

A 4

B 3

C 2

D 1

Answer: D

Explanation:
The cyclicity of 13 is 4. Now 400 is perfectly divisible by 4. Hence units digit of 13^{400} is equivalent to 13^4

which is equal to 1.

Hence, the correct answer is Option D

09. **If a* = k denotes that k is the remainder when a is divided by 7, then 100* =**

A 1

B 2

C 5

D 6

Answer: B

Explanation:
Given, a* = k denotes that k is the remainder when a is divided by 7.

Dividing 100 by 7 gives remainder 2.

Similarly, 100* = 2

Hence, the correct answer is Option B

10. **For two statements p, q, it is given that $p \to ((\sim p) \vee q)$ is false, then the truth values of p and q are respectively**

A F, T

B F, F

C T, T

D T, F

Answer: D

11. **Let p, q be two statements. Then the statement $(\sim p) \vee (p \wedge q)$ is equivalent to**

A $q \Leftrightarrow p$

B $p \Rightarrow q$

C $q \Rightarrow p$

D $p \Rightarrow \sim q$

Answer: B

12. **If $1 \leq a \leq 100$ and $A = \{a \mid gcd(a, 100) = 1\}$, then the number of elements in A is**

A 25

B 16

C 40

D 20

Answer: C

13. **Let** $f(x) = \begin{cases} x & if\ x\epsilon Q \\ 1 - x & if\ x\epsilon R - Q \end{cases}$

where Q is the set of all rational numbers. Then f is

A one-one only

B onto only

C one-one and onto

D neither one-one nor onto

Answer: C

14. **Suppose A and B are two sets. Then a set, among the following, which is not cqual to $A \cup B$, in general, is**

A $(A - B) \cup (B - A) \cup (A \cap B)$

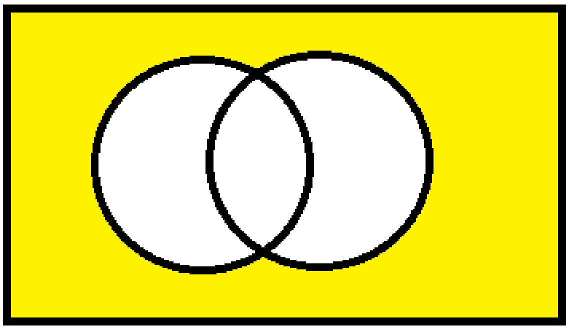
B $(A^c \cap B^c)$

C $(A - B) \cup B$

D $A \cup (B - A)$

Answer: B

Explanation:



$A^C\ intersection\ B^C$

15. **If the lines $3x - ky + 4 = 0$ and $4x + y + 2 = 0$ are perpendicular to each other, then $k^2 - 12k + 4 =$**

A 0

B 4

C 8

D 12

Answer: B

Explanation:

Slope of the line $ax + by + c = 0$ is $\frac{-a}{b}$

Slope of the line $3x - ky + 4 = 0$ is $\frac{-3}{-k} = \frac{3}{k}$

Slope of the line $4x + y + 2 = 0$ is $\frac{-4}{1} = -4$

The product of slope of two lines perpendicular to each other is -1.

$\Rightarrow \frac{3}{k} \times (-4) = -1$

$\Rightarrow k = 12$

$\therefore k^2 - 12k + 4 = 12^2 - 12(12) + 4 = 4$

Hence, the correct answer is Option B

16. **The length of the line segment intercepted between the axes by the line joining (6, -4) and (-3, 8) is**

A 4

B 5

C 6

D 7

Answer: B

17. $\sin 120^\circ \cos 60^\circ \cot 30^\circ \operatorname{cosec}^2 30^\circ =$

A 0

B 3

C -1

D $\frac{1}{2}$

Answer: B

Explanation:

$\sin 120^\circ \ \cos 60^\circ \ \cot 30^\circ \ \operatorname{cosec}^2 30^\circ \ = \sin (90 + 30^\circ) \cos 60^\circ \ \cot 30^\circ \ \operatorname{cosec}^2 30^\circ$

$= \cos 30^\circ \ \ \cos 60^\circ \ \cot 30^\circ \ \operatorname{cosec}^2 30^\circ$

$= \frac{\sqrt{3}}{2} \times \frac{1}{2} \times \sqrt{3} \times 4$

$= 3$

Hence, the correct answer is Option B

18. $\tan \theta = \frac{5}{12} \Rightarrow \frac{5 \sin \theta + 4 \cos \theta}{4 \sin \theta + 5 \cos \theta} =$

A $\frac{73}{80}$

B $\frac{80}{73}$

C $\frac{7}{80}$

D $\frac{3}{80}$

Answer: A

19. $4 \cos \theta \sin^3 \theta - 4 \sin \theta \cos^3 \theta =$

A 0

B 1

C $\sin 4\theta$

D $-\sin 4\theta$

Answer: D

Explanation:

$4\cos\theta\ \sin^3\theta\ - 4\ \sin\theta\ \cos^3\theta\ = 4\ \sin\theta\ \cos\theta\ \left(\sin^2\theta\ - \cos^2\theta\ \right) =$
$-4\ \sin\theta\ \cos\theta\ \left(\cos^2\theta\ - \sin^2\theta\ \right) = -4\ \sin 2\theta\ \cos 2\theta\ = -\sin 4\theta$

20. **A pole subtends angles $30°, 45°, 60°$ respectively at points A, B and C all lying on a horizontal line through the foot of the pole, Then $\frac{AB}{BC} =$**

A $\frac{1}{\sqrt{3}}$

B $\sqrt{3} + 1$

C $\sqrt{3}$

D $\sqrt{3} - 1$

Answer: C

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21. $x^4 - 4x^3 + 6x^2 - 4x + 1 = 0$**(x ≠ 0)** $\Rightarrow x + \frac{1}{x} =$

A $\frac{1}{2}$

B 2

C $\frac{5}{2}$

D $\frac{3}{2}$

Answer: B

22. **If x - 7 is a factor of the polynomial f(x), then a factor of $f(2x^2 - 1)$ among the following is**

A x - 1

B x - 2

C x + 1

D x + 2

Answer: B

23. **The remainder obtained when $1! + 2! + 3! + ... + (2014)!$ is divided by 7 is**

A 3

B 4

C 5

D 6

Answer: C

Explanation:
The remainder will actually depend on the sum from 1!+2!+3!+4!+...........+6! because after that every factorial will be a multiple of 7. This sums up to 873. Hence the sum will be 7m+873. Now 873 when divided by 7, has remainder 5. Hence the remainder will be 5.

24. $\sqrt{(x+1)(x+2)(x+3)(x+4)+1} =$

A $\pm(x^2+5x+4)$

B $\pm(x^2+5x+5)$

C $\pm(x^2+5x+6)$

D $\pm(x^2+6x+5)$

Answer: B

25. **The sum of seven consecutive even integers is s. Then, in terms of s, the greatest of these integers is**

A $\frac{s+20}{5}$

B $\frac{s+72}{9}$

C $\frac{s+42}{7}$

D $\frac{s+30}{6}$

Answer: C

Explanation:
Let the seven even numbers be 2a, 2a+2, 2a+4, 2a+6,2a+8,2a+10,2a+12

There sum is 14a+42=s or 14(a+3)=s or a= s/14-3

Substitute for a in 2a+12 we get s+42/7

hope it helps. Thanks

26. **The maximum value of the expression** $2+8x-x^2$ **is**

A 16

B 17

C 18

D 19

Answer: C

Explanation:
After differentiating and equating to 0,

8 - 2x = 0

$\Rightarrow$ 2x = 8

$\Rightarrow$ x = 4

$\therefore$ The maximum value of expression occurs at x = 4.

When x = 4,

$$2+8x-x^2=2+8\left(4\right)-4^2=2+32-16=18$$

Hence, the correct answer is Option C

27. $\frac{b}{a}=\frac{c}{b}=\frac{d}{c}\Rightarrow(a-c)^2+(c-b)^2+(b-d)^2-(d-a)^2=$

A 1

B 0

C -1

D 2

Answer: B

28. **The sum of first fifty odd natural numbers is**

A 2500

B 625

C 10000

D 1600

Answer: A

Explanation:
Sum of first 50 odd natural numbers can be taken as an AP with first term a=1, common difference=2 and n=50

$\frac{50}{2}\left[2\times\ 1+49\times\ 2\right]=25\left[100\right]=2500$

29. **The coefficient of x^3 in the expansion of $\left(x^2-\frac{1}{x^3}\right)^9$ is**

A $^9C_3$

B $-{}^9C_3$

C ${}^9C_5$

D $-{}^9C_4$

Answer: B

30. **The coefficient of middle term in the expansion of** $(1+x)^{40}$ **is**

A $\frac{1.3.5...39}{20!}2^{20}$

B $\frac{1.3.5...39}{20!}$

C $\frac{21.22...40}{20!}$

D $40!2^{20}$

Answer: C

Explanation:

The middle term will be the 21st term whose coefficient will be ${}^{40}C20 = \frac{40!}{20!20!} = \frac{21\times 22\times 23.....\times 40}{20!}$

31. $\begin{bmatrix} 2 & 16 \\ -8 & 0 \end{bmatrix} = \begin{bmatrix} a & b^2 \\ c^3 & 0 \end{bmatrix}, c < 0, b < 0 \Rightarrow 3a+b+c =$

A 2

B -2

C 4

D 0

Answer: D

32. **If A, B are two matrices such that AB = A, BA = B, then** A^2+B^2 **=**

A A + B

B A - B

C 2A + B

D A + 2B

Answer: A

Explanation:
AB=A and BA=B

Hence A and B are identity matrices

The square of an identity matrix is the matrix itself

Hence $A^2 = A \ and \ B^2 = B$

So $A^2 + B^2 = A + B$

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33. $\lim_{x \to 0} \frac{\tan x}{x^o} =$

A $\frac{180}{\pi}$

B $\frac{\pi}{180}$

C 1

D -1

Answer: A

Explanation:
Converting x degrees to radian. $180^o = \pi^c \ \ or \ x^o = \frac{\pi}{180} x^c$

$Now \ \lim \ x \longrightarrow 0 \ \frac{\tan x}{x} = 1$

Hence $Now \ \lim \ x \longrightarrow 0 \ \frac{\tan x}{x} \left(\frac{180}{\pi} \right) = \frac{180}{\pi}$

34. $x = \sqrt{x+y} \Rightarrow \frac{dy}{dx} =$

A 1 - x

B 1 + x

C 1 - 2x

D 2x - 1

Answer: D

Explanation:
Given, $x = \sqrt{x+y}$

$\Rightarrow \ x^2 = \ x + y$

$\Rightarrow \ y = x^2 - x$

$\Rightarrow \ \frac{dy}{dx} = 2x - 1$

Hence, the correct answer is Option D

35. **In a $\triangle$ABC, D, E, F are the mid points of the sides AB, BC and CA respectively. If AB = 8 cm, BC = 15 cm and AC = 12 om, then DE + EF + FD =**

A 16.5 cm

B 17.5 cm

C 25 cm

D 35 cm

Answer: B

Explanation:
From mid-point theorem of triangles,we can say that DE||CA and DE=1/2(CA)=6

Similarly EF||AB and EF=1/2(AB)=4

And FD||BC and FD=1/2(BC)=7.5

EF+FD+DE=6+4+7.5=17.5

36. **A, B, C are three points on the circumference of a circle with centre O. If, in $\triangle ABC. \angle B = 60°$ and $\angle C = 70°$, then $\angle BOC =$**

A $100°$

B $120°$

C $90°$

D $80°$

Answer: A

37. **If P, Q, R, S are the mid points of the sides of a quadrilateral ABCD, then the quadrilateral PQRS is a**

A Square

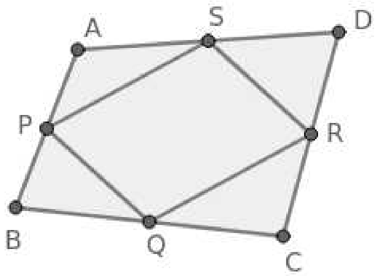
B Parallelogram

C Rectangle

D Rhombus

Answer: B

Explanation:



Quadrilateral formed after joining the mid-points of another quadrilateral is always a parallelogram.

∴ Quadrilateral PQRS is a parallelogram.

Hence, the correct answer is Option B

38. **The points A(3, -5) and B(-5, 4) are given. If C is a point such that $\frac{AC}{CB} = 2$, then the coordinates of C are**

A $\left(\frac{7}{3}, 1\right)$

B $\left(\frac{-7}{3}, 1\right)$

C $\left(\frac{7}{3}, -1\right)$

D $\left(\frac{-7}{3}, -1\right)$

Answer: B

Explanation:
$\frac{AC}{CB} = 2 \ or \ AC = 2CB$

Hence C divides AB in the ratio of 2:1

Using the proportionality theorem of coordinate geometry

$x = \frac{2\times(-5)+1\times3}{3} = -\frac{7}{3}$

$x = \frac{2\times(4)+1\times(-5)}{3} = 1$

Hence the coordinates are (7/3,1)

39. **A (4, 2), B (6, 5) and C (1, 4) are the vertices of a $\triangle$ABC. The median from A meets the side BC at D. Then $2AD^2 =$**

A 13

B 14

C 15

D 16

Answer: A

Explanation:
D is the midpoint of BC. Hence coordinates of D is (3.5,4.5)

Now applying distance formula , AD is $\sqrt{(0.5)^2 + (2.5)^2} = \sqrt{0.25 + 6.25} = \sqrt{6.5}$

$2AD^2 = 2 \times \ 6.5 = 13$

40. **The mean of the distribution given below is**

x	10-20	20-30	30-40	40-50
Frequency	5	10	7	8

A 30

B 31

C 32

D 33

Answer: B

Explanation:
We take the mid-value of each interval for the calculation of mean of the given distribution.

$\therefore$ Mean = $\frac{(15\times5)+(25\times10)+(35\times7)+(45\times8)}{5+10+7+8} = \frac{930}{30} = 31$

Hence, the correct answer is Option B

41. **For a given data, if the mean is 60 and the mode is 66, then the median is**

A 63

B 64

C 60

D 62

Answer: D

Explanation:
Given, Mean = 60 and Mode = 66

Using the relation, Mode = 3Median - 2Mean

66 = 3Median - 2(60)

$\Rightarrow$ 3Median = 66 + 120

$\Rightarrow$ 3Median = 186

$\Rightarrow$ Median = 62

Hence, the correct answer is Option D

42. **The mode of the following data is**
6, 9, 13, 10, 16, 13, 13, 14, 15, 11, 13, 12, 14

A 11

B 12

C 13

D 14

Answer: C

Explanation:
A mode is the observation with the most frequency or that observation that appears the most number of times. Hence 13 is the answer as it appears 4 times. more than anyone else

43. **If σ is the standard deviation of $x_1, x_2, ..., x_n$, then the standard deviation of $9 + 3x_1, 9 + 3x_2, ..., 9 + 3x_n$ is**

A $3\sigma - 3$

B $\sqrt{9\sigma^2 + 3}$

C 3σ

D $3\sigma + 9$

Answer: C

44. **The vanance of first n even natural numbers is**

A $\frac{n^2-1}{3}$

B $\frac{n^2-1}{6}$

C $\frac{n^2-1}{12}$

D $\frac{n^2+1}{3}$

Answer: A

45. **The mean of first n odd natural numbers is**

A n - 1

B n + 1

C n + 2

D n

Answer: D

46. **A number is selected at random from the first 80 natural numbers. The probability thatit is divisible by 4 or 6 is**

A $\frac{23}{80}$

B $\frac{29}{80}$

C $\frac{27}{80}$

D $\frac{33}{80}$

Answer: C

Explanation:
Numbers divisible by 4- 20

Numbers divisible by 6-13

Numbers divisible by 4 or 6- 6

Total numbers 20+13-6=27

Probability =27/80

47. **Two fair dice are rolled. The probability that the sum of the numbers on the faces shown is 8 is**

A $\frac{5}{36}$

B $\frac{1}{6}$

C $\frac{7}{36}$

D $\frac{1}{9}$

Answer: A

Explanation:
The possible outcomes that the sum of the numbers on the faces is 8 are (6,2), (2,6), (3,5), (5,3), (4,4).

Number of possible outcomes = 5

Total number of outcomes = 36

$\therefore$ Required probability = $\frac{5}{36}$

Hence, the correct answer is Option A

48. **The probability that either of the events A and B to happen is 0.6 and the probability that both of them to happen is 0.2. Then P(A') + P(B') =**
(Here A' is the complementary event of A.)

A 0.4

B 0.75

C 0.8

D 1.2

Answer: D

49. **Suppose f(x) = (x - 2) (x - 5) (x - 7).**

If a number α is chosen from {1, 3, 4, 5, 6, 7, 8, 9, 10} randomly, the probability that it satisfies the equation $f(\alpha) = 0$,is

A $\frac{1}{3}$

B $\frac{2}{5}$

C $\frac{3}{7}$

D $\frac{2}{9}$

Answer: D

Communication Ability

Instructions [150 - 155]

Choose the correct meaning for the word given :

50. **radical**

A extreme

B red

C colourful

D slow

Answer: A

51. **tether**

A teeth

B together

C restrain

D free

Answer: C

52. **synergy**

A combined size

B joined effort

C related parts

D organized finances

Answer: B

53. **pervade**

A conquer

B escape

C spread through

D convince

Answer: C

54. **nascent**

A smelly

B fragrant

C immune

D incipient

Answer: D

55. **vacuous**

A abandon

B vacate

C unavailable

D expressionless

Answer: D

Instructions [156 - 159]

Fill in the blank choosing the correct word :

56. **The rains .... the fields, washing away the crops.**

A stormed

B inundated

C blew

D covered

Answer: B

57. **Due to the ongoing controversy the political situation in the state is ....**

A upbeat

B cosy

C turbulent

D sublime

Answer: C

58. **England was a great .... power in the nineteenth century.**

A merchant

B army

C mercantile

D navy

Answer: C

59. **The .... mule would not pull the farmer's plow.**

A rigid

B sturdy

C rugged

D stubbom

Answer: D

Instructions [160 - 169]

Choose the correct answer:

60. **The process of reviewing the performance of employees periodically is called**

A performance management

B employee review

C performance appraisal

D employee confidential report

Answer: C

61. **The interview conducted in a situation not quite pleasant or comfortable to the candidate is called**

A unstructured interview

B depth interview

C stress interview

D distress interview

Answer: C

62. **The medium of outdoor poster in which printed ad message is displayed is called**

A cutouts

B POP

C exhibit

D bill board

Answer: D

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63. **A market which is dominated by a few suppliers is known as**

A perfect market

B buyer's market

C oligopoly market

D monopolist market

Answer: C

64. **PERT is**

A Programme Evaluation and Review Technique

B Programme Education and Review Teaching

C Programme Enlightment and Review Technique

D Progress Evaluation and Review Timing

Answer: A

65. **Which of the following is used for modulation and demodulation ?**

A Modem

B Protocols

C Gateway

D Multiplexer

Answer: A

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66. **Linkage between CPU and users is provided by**

A storage

B control unit

C penpheral devices

D software

Answer: C

67. **In a computer system, which device is functionally opposite of a keyboard ?**

A mouse

B track ball

C printer

D pen drive

Answer: C

68. **The first mechanical computer designed by Charles Babbage was called**

A Abacus

B Processor

C Calculator

D Analytical Engine

Answer: D

69. **Which of the following is an example of non-volatile memory ?**

A ROM

B VLSI

C LSI

D RAM

Answer: A

Instructions [170 - 176]

Choose the correct answer ;

70. **A: "There, that's what you looked like when you were a month old."**
B: "How awful!"
In this conversation 'B'

A is pleased

B is disappointed

C is unhappy

D thinks he looked ugly

Answer: D

71. **A: "Could I borrow some money from you ?"**
B: "What do you need it for?"
The conversation implies that 'B'

A is dodging the issue

B is angry

C does not want to give money

D wants to know why 'A' needs money

Answer: D

72. **"It was a knockout. Umesh saw stars in his eyes."**
The speaker implies that Umesh. |

A is exuberant

B is romantic

C is dreaming

D has fallen unconscious

Answer: D

73. **The active form of the sentence 'e-mails have been written by her' is**

A she has written e-mails

B she had written e-mails

C write e-mails to her

D she sent e-mails

Answer: A

74. **A: "I've got anew job"**
B: "Great! that should open doors for you."
'B' implies that

A 'A' can get more opportunities.

B The job will be done outdoors.

C The job will fetch a lot of money.

D ‘A’ will be able to work well,

Answer: A

75. **A: “There is a lot of disunity among the people.”**
B: “I agree. Unity is the crying need of the hour.”
‘B’ implies that

A there is no need for uniy among the people.

B there is an urgent need for unity among the people.

C a plan may be made for achieving unity.

D unity may be achieved by crying for it, hour by hour.

Answer: B

76. **A: “I want to train myself in yoga practices.”**
B: “I want to follow suit.”
‘B’ implies

A that he would putoon a suit.

B that he would follow anyone wearing a suit.

C that he would imitate ‘A’.

D that it does not suit his temperament to follow anyone.

Answer: C

Instructions [177 - 184]

Fill in the blanks with the appropriate phrase/verb/preposition:

77. **Manish had a poor salary but he didn't need much to ....**

A live up

B live on

C live after

D live in

Answer: B

78. **Anthony .... his wife to tell her that he would reach home late.**

A called up

B called on

C called at

D called away

Answer: A

79. **He has to .... with the eccentricity of his boss.**

A put on

B put up

C put away

D put in

Answer: B

80. **Great people achieve what the others only dream ....**

A by

B of

C with

D out

Answer: B

81. **She saved the child .... drowning.**

A off

B from

C for

D through

Answer: B

82. **She'll be fearful .... that.**

A on

B of

C with

D to

Answer: B

83. **Being very tired .... studying.**

A impinges

B impeaches

C inhibits

D inculeates

Answer: C

84. **They .... cultivating the land for twenty years when they moved to the city.**

A had been

B are

C is

D have been

Answer: A

Instructions [185 - 189]

Read the following passage and answer the questions:

After two decades of growing student enrollments and economic prosperity, business schools in the USA have started to face harder times. Only Havard's MBA school has shown a substantial increase in enrollment in recent years, Both Princeton and Stanford have seen decreases in their entrollments. Since 1990, the number of people receiving MBA degrees has dropped about 3 percent to 75,000 and the trend of lower enrollment rate is expected to continue.

There are two factors causing this decrease in students secking MBA degree. The first one is that many graduates of four year colleges are finding that an MBA degree does not guarantee a plush job on Wall Street or in other financial districts of major American cities. Many of the entry level management jobs are going to students graduating with Master of Arts degrees in English and the humanities as well as those holding MBA degrees. Students have asked the question, "Is an MBA degree rcally what I need to be best prepared for getting a good job ?" The second major factor has been the cutting of American payrolls and the lower number of entry-level jobs being offered. Business needs are changing, and MBA schools are struggling to meet the new demands.

85. **Which of the following business schools has not shown a decrease in enrollment ?**

A Princeton

B Harvard

C Stanford

D Yale

Answer: B

86. **What is the duration of an MBA degree ?**

A 4 years

B 3 years

C 2 years

D not mentioned in the text

Answer: D

87. **What are the two causes of declining business school enrollments ?**

A Lack of necessity for an MBA and an economic recession.

B Low salary and foreign competition.

C Declining population and low education standard.

D Fewer MBA schools and higher tuition fees.

Answer: A

88. **Which are the degrees preferred along with MBA for entry-level management jobs ?**

A Post Graduation in Linguistics

B Graduation in humanities

C Masters programme in Arts and Literature

D Master in English and Humanities

Answer: D

89. **What should be done by business schools to change the situation ?**

A Charge lower fee

B Examine the changing needs of business

C Change the curriculum

D Improve placement procedure

Answer: B

Instructions [190 - 194]

Read the following passage and answer the questions:

More businesses are addressing social issucs through philanthropy. Companies donate a portion oftheir revenues to charitics or a specific social cause. Education is known to be the favourite object for philanthropy in which 75 percent of companies are participating. Although the donations will help a good cause, many companies use philanthropy primarily to improve their reputation or get a tax deduction. Philanthropy is not limited to the mature markets in the West. In emerging markets philanthropy is even more popular. Asia's millionaires committed 12 percent of their wealth for social causes. While millianaires in North America only contribute § percent and those in Europe 5 percent.

Although philanthropy helps society, we should never over estimate its sociocultural impact. Recent growth in philanthropy is driven by the changes in the society. Even in a recession, 75 percent of Americansstill donate to a social cause.

But philanthropy does not stimulate transformation in the society, Transformation in the society drives philanthropy. That is why addressing social issues with philanthropic activities will have a rather short-term impact.

A more advanced form of addressing social challenges is cause marketing — a practice where companics support a specific cause through their marketing activities.

90. **Why do companies set aside money in their budget for charities ?**

A It helps to reduce their tax liabilities.

B People want to help others.

C Companics do not want to attract attention.

D It makes people in the company happy.

Answer: A

91. **What is the change that is coming about now ?**

A Marketing has become easier.

B Companies have started earning more.

C There are more advertising companies.

D There is a growth in social awareness.

Answer: D

92. **What is the most favourite area for donations from companies ?**

A Healthcare

B Education

C Social Ethics

D Public Relations

Answer: B

93. **What according to the author, will have only a limited impact on the transformation of society ?**

A Philanthropy

B Social change

C Recession

D Marketing strategies

Answer: A

94. **What is the main idea of this passage ?**

A Philanthropy focuses only on education.

B Western countries spend more than others on philanthropy.

C Cause Marketing is a better form of marketing.

D Companies donate some part of their income to charities.

Answer: C

Instructions [195 - 199]

Read the following passage and answer the questions:

To many people growing old seems like the end game in chess : life winding down in a series of small moves with lesser pieces. As I age, I have discovered this is nottrue. I am not an elderly king stripped of my powers, reduced to a ragtail army of pawns. Mylife is not a defensive struggle of restricted options. Growing old is a game of verve and imagination and excitement. The outcome is not now a matter of strength, although that still remains, but of faith and courage, hope and wisdom. The aging game is a sport for which childhood and youth and maturity are no more than a preparation. Its scope comes a surprise. It expands my life at a time when I expected it to diminish. It demands an excellence that no longer seemed necessary. It asks me to surpass what I did at the peak of my powers. Age will not accept second best. In the aging game I must be all ever I was and am yet to be. What has gone before is no more than a learmming period. A breaking in. Age is the combat for which I was trained. Now I must take this person I have become and make each new day special, I must make good on the promise of every dawn I am privileged to see. Life goes from a minor to a major key. The game builds to a climax. Every move assumes importance. One feels like a virtuoso, the gifts we have been given, the powers that empower us, the marvels that make us marvellous, are evident as never before. The truth is that we have lost nothing. The problem is not that I am less than I was when young, it is that J am not more. It is past time to become my own person.

95. **What does growing old mean to many people ?**

A the end of the challenges

B mental degeneration

C lack of activity

D boredom

Answer: A

96. **What does aging mean to the author ?**

A to be negative

B to be positive

C to laze around

D to be depressed

Answer: B

97. **What should aging lead to ?**

A perfection

B death

C illness

D a marvellous existence

Answer: A

98. **Why is the 'aging game' referred to as a 'sport' ?**

A Old people play games.

B Old age makes one young in spirit.

C Problems of old age have to be overcome.

D It is a game in which one loses or wins.

Answer: D

99. **What does childhood and maturity prepare are for ?**

A aging

B to face old age with hope and wisdom

C to rest in old age

D to be prepared for illness in old age

Answer: B